EWB AWS VPC Peering — Project Report

Student name: vivin francis
Batch: batch 1
Project: AWS VPC Peering
AWS Region: Asia Pacific (Sydney), ap-southeast-2
Completion date: oct 30

1. Objective

Build private connectivity between two non-overlapping VPCs using AWS VPC Peering. Keep the application instances private and use one public bastion host in VPC A for administration.

2. Architecture

- VPC A: 10.10.0.0/16
  - Public subnet: 10.10.1.0/24 — Bastion 10.10.1.233
  - Private subnet: 10.10.2.0/24 — App-A 10.10.2.184
- VPC B: 10.20.0.0/16
  - Private subnet: 10.20.1.0/24 — App-B 10.20.1.31
- Peering connection: ewb-a-b-peer — Active
- Internet Gateway: ewb-igw-a

Insert architecture diagram and AWS resource-ID inventory here.

3. Routing and security

The public route table in VPC A contains the local route, default route to the Internet Gateway, and 10.20.0.0/16 to the peering connection. The VPC A private route table contains 10.20.0.0/16 to peering. The VPC B private route table contains 10.10.0.0/16 to peering. These routes provide forward and return paths.

SSH was restricted to the bastion private IP for private instances. ICMP was allowed only from the relevant lab peer/test source addresses. The bastion SSH rule should be restricted to the administrator's current public IPv4 /32.

Insert screenshots of the three route tables and security groups here.

4. Validation results actually confirmed

Test	Result
Peeringstate	Active
LaptoptoBastionSSH	Successful
BastiontoApp-Aping	Successful;0%packetlossreported
BastiontoApp-Bping	Successful;0%packetlossreported

Test	Result
BastiontoApp-ATCPport22	Reachable;Pythonsocketreturned0
Laptop to App-A through Bastion	Successful
App-AtoApp-Bping	Successful

Insert original terminal screenshots for each result. The table above records results reported during the lab; attach original evidence rather than recreating or editing terminal output.

5. Required evidence still to complete

Do not mark these as completed until you have run and captured them: - App-B to App-A reverse ping. - SSH from laptop through Bastion to App-B. - Controlled ICMP failure and recovery: remove only the App-B inbound ICMP rule sourced from App-A, test that App-A ping fails, restore the rule, and show ping recovery. - Baseline failure before peering, if it was not captured before enabling peering. Do not present a later test as a pre-peering baseline. - Confirm App-A and App-B have no public IPv4 addresses. - Capture resource IDs and cleanup evidence.

6. Learning outcomes

This lab demonstrated VPC CIDR planning, subnetting, route-table association, Internet Gateway use, VPC Peering, return routing, security-group filtering, SSH ProxyJump, and private-IP connectivity testing.

7. Cleanup

After evidence is saved, terminate the lab EC2 instances and check for leftover EBS volumes/snapshots and Elastic IPs. Remove the peering connection, security groups, subnets, route tables, Internet Gateway, and VPCs in dependency order. Verify no lab resources remain in ap-southeast-2 and review billing.